

Payments Operations & Revenue Reconciliation

Project brief: business context, rules, data and analysis plan

Project overview

Tarang Pay is a payment gateway for Indian online businesses. Merchants use it to accept UPI, card, netbanking and wallet payments. Tarang Pay collects the money, keeps a fee, and pays the rest out to the merchant on the next working day.

Tarang Pay is a fictional company. Its banks, merchants and data were created for this project, and every rule below is treated as true.

Business problem

The payments operations team noticed that payment volume nearly doubled this year, from ₹8.9 crore to ₹17 crore a month, but fee revenue grew only by a third. At the same time, numbers that didn't match between systems (reconciliation differences), a spike in refunds and late payouts to merchants made leadership unsure which figures to trust before the investor update. Three teams see the problem differently:

- **Finance:** fee revenue in the books is lower than the volume should produce.
- **Operations:** merchants complain that some payments reach them late, or never.
- **CFO:** refunds spiked in one week of August, and it isn't clear whether something broke.

Main business question. Payment volume doubled this year, but revenue didn't. **Which one fix should get October's engineering sprint?**

Engineering can give payments operations only one sprint in October, so the analysis has to end in a single choice, with rupees attached. On the way, it answers three supporting questions: which numbers can be trusted, where money is going missing, and what to fix first.

Business rules (treated as true)

Orders and attempts

- A customer places an order with a merchant. Each try to pay for it is an attempt, with its own `txn_id`.
- If an attempt fails, the customer may retry. That creates a new attempt with the same `order_id` and the next `attempt_no`.
- An order is successful if any of its attempts succeeded.

Payment status

- An attempt ends as success or failed. Until then it is pending, which must resolve within 48 hours.
- `status_updated_at` records when the final status reached Tarang Pay.

Settlement

- A successful payment is paid out to the merchant on the next working day (T+1). Working days exclude Sundays, the second and fourth Saturdays, and the bank holidays in `calendar.csv`.
- Day T is the day the payment's final success status reached Tarang Pay. A status that arrives at or after 23:00 IST counts toward the next day.
- Merchants whose KYC is `on_hold` are not paid out until it clears. That is correct behaviour, not an error.

Fees and GST

- $\text{Fee} = \text{payment amount} \times \text{the rate in pricing.csv for that merchant and method, on the payment date.}$
- UPI carries a 0% fee for every merchant.
- GST at 18% is charged on the fee, not on the payment amount.
- The fee and GST are deducted at settlement. Finance books fee revenue (ex-GST) in `revenue_ledger.csv.gz`, one entry per settlement batch, on the settlement date.

Refunds

- A merchant can refund a successful payment, fully or partly, up to 90 days after it.
- Tarang Pay keeps its fee when a payment is refunded.

All money is in rupees. All times are IST. The data runs from 1 January to 31 August 2026.

The data

File	One row per	Rows
merchants.csv	merchant	2,010
pricing.csv	merchant × method × period a rate applies	11,840
transactions.csv.gz	payment attempt	506,062
settlements.csv.gz	payment paid out to a merchant	445,917
revenue_ledger.csv.gz	fee entry in finance's books (one per settlement batch)	72,876
refunds.csv	refund	13,013
calendar.csv	day, with a working-day flag and holiday name	243
monthly_report.csv	month, as currently reported to leadership	8

Columns

Column	Meaning
order_id, txn_id, attempt_no	Order, attempt, and which try it was (1 = first)
created_at	When the attempt started
status, status_updated_at	Final status, and when it reached Tarang Pay (blank if still pending)
failure_reason	For failed attempts: customer_dropped, insufficient_funds, bank_declined, bank_timeout, technical_error
method, bank	Payment method, and the customer's bank
settlement_id	Settlement batch: one per merchant per settlement date
payment_amount, fee_charged, gst_charged, settled_amount	The payment, what Tarang Pay kept, and what the merchant received
fee_booked, booked_date	Fee revenue as finance recorded it (ex-GST)
fee_rate_pct, valid_from, valid_to, plan	Agreed fee rate and when it applies (blank valid_to = still in force)
category, size_tier, joined_date	Merchant details
kyc_status, kyc_hold_since	KYC state, and the date a hold started

Analysis plan

The analysis is organised in six parts, each answering one piece of the business problem.

Part 1 • Data validation

Before any analysis, confirm that the data means what it says.

- Profile each file: completeness, ranges, duplicates, impossible values.
- Check across files. Does every settled payment have a successful attempt? Does every settlement batch have a ledger entry? Does every fee match the pricing table? Does every refund belong to a successful payment, and is any refund larger than its payment or later than 90 days?
- Separate real errors from things that only look wrong.

Each finding goes into an investigation log:

What was noticed	Rows and ₹ affected	What probably happened	How it was handled	Why that handling
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Every finding is handled in one of four ways: corrected, excluded, flagged but kept, or left alone. Financial data is not changed without proof. The log ends with the questions engineering or finance would need to answer before these numbers can be fully trusted.

Part 2 · Business performance

- One row per month: attempts, orders, success rate, TPV (value of successful payments), average payment value, take rate (fee revenue ÷ TPV), refund rate, active merchants.
- Choose a definition of success rate (orders or attempts), and show what the other one would give.
- Where payments fail: by method, bank, hour of day and merchant category. Which failures can Tarang Pay fix, and which are the customer's choice?
- Compare with `monthly_report.csv`: which reported numbers are wrong, by how much, and why?
- Take rate fell this year. Is that bad news?

Key pitfall: timing. Money doesn't move on the day a payment happens. A settlement or a ledger entry on a given day belongs to payments from an earlier working day, and a refund belongs to a payment from days or weeks earlier. Comparing same-day totals across these files creates gaps that aren't real and hides some that are.

Part 3 · Where the money goes missing

Follow successful payments through the chain: successful payment → paid out to the merchant → fee booked by finance. Report the settlement completion rate (successful payments that were paid out) and the on-time settlement rate (paid out on the next working day after the payment), by month. Then:

- The settlement gap. Split late and missing payouts into their causes. For each: rows, ₹, whether it is expected behaviour, a process problem or a data problem, and who should fix it.
- Fees. Compare the fee that should have been charged (from the pricing table) with the fee actually charged. Where they differ: ₹, merchants, and when it started.
- Finance's books. Compare fees charged at settlement with fees booked. Is there a gap, and is it the same problem as the one above?

A difference is called potential leakage until the evidence proves money was lost, with a note on what evidence would prove it.

Part 4 · Merchants

- Concentration: what share of TPV and of fee revenue comes from the top 10% of merchants?
- Retention: for merchants who joined each month from January to June, what share were still taking payments in each later month?
- The link: do merchants whose payments often fail in their first month leave sooner? What else could explain it?
- Call list: the ten merchants the account team should call this week, with one line on why for each.

Part 5 · One engineering sprint

Engineering can give payments operations one sprint in October. The options:

- A. Fix the integration with the bank that sends late status updates.
- B. Fix the pricing configuration, and recover the fees that weren't charged.
- C. Build a daily automated reconciliation that alerts when payments, settlements and the ledger don't match.

For each option: the ₹ impact per month (or the risk it removes), the assumptions, and the confidence level. Choose one, and give:

- The evidence that would prove the choice wrong.
- One headline number for leadership, and why that one.
- One metric for the monthly investor deck (success rate, TPV or net take rate), argued in three sentences.

Part 6 · One chart

One chart, not a dashboard, that the Head of Payments could read in ten seconds and know what to do. Underneath it, one sentence saying what it tells them to do. If that sentence can't be written, it's the wrong chart.

Approach

- Open questions are written down, each with the assumption made in the meantime, instead of guessing silently.
- No machine learning. Every question here can be answered with careful joins, grouping and business logic, so a model would add complexity without adding insight.
- Every finding has a rupee amount, not just a percentage.

Deliverables

- Analysis notebook: a Python notebook that reruns top to bottom from the raw files.
- Memo: one page with the recommendation first, evidence second and caveats last, plus one page for the investigation log and open questions.
- Charts, including the one chart for the Head of Payments.

Quality standards

Area	What good looks like	What to avoid
Data judgement	The right handling for each problem, with the ₹ attached	Deleting rows silently, or correcting financial data without proof
Grain	Orders vs attempts, and payment date vs settlement date, used correctly	Joins that duplicate rows; same-day comparisons across files
Root cause	The settlement gap split into separate causes with owners	"There is a 3% gap"
Money	Every finding has a rupee amount	Percentages only
Decision	One sprint chosen, with what would change the decision	Recommending everything
Communication	A leader understands the memo without opening the notebook	The notebook is the answer